

ENTELSO OPERATIONS MANUAL

Intelligent Inventory & Asset Tracking System for Technical Equipment

This comprehensive documentation serves as the official operational manual for the Entelso ecosystem. It is divided into two specialized guides tailored for different operational roles:

- 1. Warehouse Administration & Dashboard Management:** Designed for supervisors, warehouse managers, and administrative personnel who oversee global equipment inventory, compliance tracking, batch uploads, and alert management via the web platform.
- 2. Field Technician & WhatsApp Bot Operations:** Designed for engineers and technicians operating on-site, enabling instant asset lookup, real-time custody transfers, geolocation verification, and incident reporting directly through WhatsApp without requiring specialized software installations.

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Target Audience:	Warehouse Supervisors & Field Technicians
System Architecture:	Node.js Backend, Supabase Storage, WhatsApp Cloud API
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PART I: WAREHOUSE ADMINISTRATION & DASHBOARD GUIDE

1. Executive KPI Monitoring & Live Analytics

The Entelso web dashboard serves as the central command hub for warehouse supervisors. Upon logging in, administrators are presented with real-time operational indicators:

- **Total Asset Fleet:** The complete tally of equipment registered in the global inventory.
- **Field Deployment Ratio:** Percentage and count of assets currently assigned to technicians or deployed in operational zones.
- **Available Stock:** Equipment currently residing in warehouse custody and ready for immediate deployment.
- **Critical Compliance Alerts:** Immediate indicators for equipment approaching calibration limits or requiring mandatory inspection tags.

2. Bulk Multi-Asset Import with Embedded Excel Photos

Entelso features an advanced ingestion engine capable of processing existing enterprise spreadsheets (such as *Equipment Register.xlsx*). Unlike standard web parsers that ignore embedded graphics, Entelso supports importing equipment records alongside their floating physical photographs.

How the Bulk Ingestion Engine Works:

Step 1 — Spreadsheet Formatting: Ensure your spreadsheet contains required columns: Asset ID / Inventory Number, Description, Brand/Model, Serial Number, and Condition.

Step 2 — Embedded Photos: Photographs inserted as floating drawing objects in Excel columns (e.g., Photo 1, Photo 2) are automatically matched to their corresponding equipment row.

Step 3 — Cloud Storage Synchronization: When ingested, physical photographs are extracted, compressed into optimized JPEG formats, and uploaded securely to Supabase Storage (*entelso-media* bucket).

Step 4 — Database Linking: Public cloud URLs are permanently bound to the asset record in the PostgreSQL database, making them immediately visible across the web dashboard and mobile WhatsApp queries.

■ **PRO TIP — EXCEL PHOTO ALIGNMENT:** When preparing spreadsheets for bulk ingestion, ensure that top-left anchor of each photograph sits cleanly within the boundary of its equipment row. If an item lacks a photograph in the sheet, its image gallery can be populated later directly from the dashboard or via WhatsApp.

3. Alert Management & Expiration Tracking

Compliance and safety are paramount when deploying technical gear such as safety lines, harnesses, and calibration analyzers. The dashboard automatically calculates expiration timelines and categorizes assets into actionable alert tiers:

- **Critical / Expired (Calibración Vencida / Tag Vencido):** The calibration date or NFC safety tag inspection has passed its due date. The asset is highlighted in red and should be quarantined from field deployment.
- **Warning / Due Soon (Calibración Próxima / Tag Próximo):** Equipment approaching its mandatory calibration or inspection expiration within the next 30 to 60 days. Supervisors should schedule maintenance windows.
- **Compliant / Available:** Equipment fully certified, calibrated, and cleared for operational assignment.

4. Asset Detailed Audit Drawer & Specifications

Clicking any equipment row in the main inventory grid opens the **Detailed Audit Drawer** on the right side of the screen. This panel provides granular control over the individual asset:

- **Photo Gallery Management:** View all synced photographs, enlarge images, or upload supplementary pictures directly from your local computer.
- **QR Code & NFC Tag Generation:** Generate high-resolution, unique QR codes for physical tagging. Technicians can scan these labels in the field to trigger instant WhatsApp queries.
- **Custody Reassignment:** Manually assign equipment to a specific engineer or transfer it between warehouse locations and operational zones.
- **Immutable Telemetry History:** Access the chronological audit log detailing every status check, location update, and transfer ever performed on the tool.

PART II: FIELD TECHNICIAN & WHATSAPP BOT GUIDE

1. Instant Field Operations via WhatsApp

Field technicians do not require specialized applications or VPN installations to interact with the Entelso inventory system. The official **Entelso WhatsApp Assistant** serves as an intelligent, real-time bridge to the core database, allowing engineers to verify equipment safety, report locations, and transfer custody in seconds.

2. WhatsApp Command Cheatsheet

To execute commands, send the formatted text directly to the Entelso WhatsApp number. The assistant accepts both English and Spanish natural variations.

Command / Action	Syntax Example	System Response & Workflow
Asset Query (Check Status)	EQ-01 or STATUS EQ-01	Returns complete asset profile: Description, brand, current condition, calibration compliance status, assigned custodian, and physical photos.
Checkout / Take Custody (Assign to Self)	CHECKOUT EQ-01 or TOMAR EQ-01	Assigns the tool to your user account, sets status to 'In Use', and logs the timestamp in the audit trail. Requires PIN verification.
Checkin / Return (Return to Warehouse)	CHECKIN EQ-01 or DEVOLVER EQ-01	Releases personal custody, resets asset location to warehouse availability, and updates compliance metrics.
Report Incident (Flag Damage)	REPORT EQ-01 broken strap or DAÑO EQ-01 cable pelado	Immediately flags the asset as 'Under Maintenance / Damaged', triggers a priority alert on the supervisor dashboard, and logs the notes.
My Inventory (List Assigned Gear)	MY ASSETS or MIS EQUIPOS	Generates an itemized summary of all tools and instruments currently assigned to your personal custody.

■■ **MANDATORY SAFETY NOTICE:** Before checking out fall-protection equipment (such as Harnesses, Safety Lines, or Lanyards), technicians **MUST** review the expiration date returned by the WhatsApp bot. If the system reports 'Calibración Vencida' or 'Tag Vencido', **DO NOT** deploy the gear in the field.

3. Geolocation Verification & Zone Tracking

When operating across expansive client facilities or remote cell sites, maintaining geographical traceability is essential. Technicians can share their live location directly through WhatsApp to geo-tag equipment deployments:

Step 1: Query or check out the equipment via WhatsApp (e.g., send CHECKOUT EQ-09).

Step 2: Tap the attachment icon (■) in WhatsApp and select **Location** → **Send Your Current Location**.

Step 3: The system binds the GPS coordinates to the asset's active telemetry log, allowing supervisors to visualize field distribution across operational zones.

4. Security PIN & 2-Factor Authentication

To prevent unauthorized asset transfers and ensure accountability, sensitive operations require personal authentication. Every authorized field engineer is assigned a unique 4-digit Security PIN and a linked WhatsApp phone number:

- **PIN Verification:** When attempting to transfer high-value instruments or report critical damage, the assistant will prompt: *'Please enter your 4-digit security PIN to confirm.'*
- **Account Identity Binding:** Commands are strictly validated against the sender's registered WhatsApp phone number. If an unrecognized number attempts to query inventory, access is denied and logged for review.

End of Official Operations Manual. For technical support or system configuration inquiries, contact the Entelso IT & Infrastructure Engineering Team.